TUGAS PEMROGRAMAN WEB LANJUT

PHP-OOP

EXPLORER

UNTITLED (WORKSPACE)

OOP-PHP

index.php

mahasiswa.php

mahasiswa.php | index.php | Extension: Java

OOP-PHP > mahasiswa.php

```php
<?php
class Mahasiswa
{
    public $nim;
    public $nama;
    public $programStudi;

    function setData($nim, $nama, $programStudi)
    {
        $this->nim = $nim;
        $this->nama = $nama;
        $this->programStudi = $programStudi;
    }

    function getData()
    {
        return [
            'nim' => $this->nim,
            'nama' => $this->nama,
            'programStudi' => $this->programStudi
        ];
    }

    // method baru
    function tampilData(){
        echo "NIM: " . $this->nim . "<br>";
        echo "Nama: " . $this->nama . "<br>";
        echo "Program Studi: " . $this->programStudi . "<br><br>";
    }
}
?>
```

EXPLORER

UNTITLED (WORKSPACE)

OOP-PHP

index.php

mahasiswa.php

mahasiswa.php | index.php | Extension: Java

OOP-PHP > index.php

```php
<?php
include "Mahasiswa.php";

$mahasiswaTI = new Mahasiswa();
$mahasiswaTI->setData('A11.2024.15812', 'Anthonius Raffael Pangaribuan', 'Teknik Informatika');
print_r($mahasiswaTI->getData());

echo "<br><br>";

// objek kedua
$mahasiswaSI = new Mahasiswa();
$mahasiswaSI->setData('A12.2000.00001', 'Latihan', 'Sistem Informasi');
print_r($mahasiswaSI->getData());

echo "<br><br>";

// objek ketiga
$mahasiswaMI = new Mahasiswa();
$mahasiswaMI->setData('A13.2000.00002', 'Latihan', 'Manajemen Informatika');
print_r($mahasiswaMI->getData());

echo "<h2>Data Mahasiswa:</h2>";

$mahasiswaTI->tampilData();
$mahasiswaSI->tampilData();
$mahasiswaMI->tampilData();
?>
```

OUTPUT

localhost/oop-php/index.php

Array ([nim] => A11.2000.00001 [nama] => Fulan [programStudi] => Teknik Informatika)

Array ([nim] => A12.2000.00001 [nama] => Junior [programStudi] => Sistem Informasi)

Array ([nim] => A13.2000.00001 [nama] => Andi [programStudi] => Manajemen Informatika)

localhost/oop-php/index.php

Array ([nim] => A11.2024.15812 [nama] => Anthonius Raffael Pangaribuan [programStudi] => Teknik Informatika)

Array ([nim] => A12.2000.00001 [nama] => Latihan [programStudi] => Sistem Informasi)

Array ([nim] => A13.2000.00002 [nama] => Latihan [programStudi] => Manajemen Informatika)

Data Mahasiswa:

NIM: A11.2024.15812
Nama: Anthonius Raffael Pangaribuan
Program Studi: Teknik Informatika

NIM: A12.2000.00001
Nama: Latihan
Program Studi: Sistem Informasi

NIM: A13.2000.00002
Nama: Latihan
Program Studi: Manajemen Informatika

LATIHAN PPT PERTEMUAN 2

LATIHAN 1

EXPLORER

UNTITLED (WORKSPACE)

lat1

Index.php

Mahasiswa.php

MataKuliah.php

Mahasiswa.php | MataKuliah.php | Index.php

lat1 > Mahasiswa.php

```php
<?php
class Mahasiswa
{
    private $nim, $nama, $matkul;

    public function setData($nim, $nama, $matkul)
    {
        $this->nim =$nim;
        $this->nama =$nama;
        $this->matkul =$matkul;
    }

    public function getData()
    {
        return[
            $this->nim,
            $this->nama,
            $this->matkul,
        ];
    }
}
```

EXPLORER

UNTITLED (WORKSPACE)

lat1

Index.php

Mahasiswa.php

MataKuliah.php

Mahasiswa.php | MataKuliah.php | Index.php

lat1 > MataKuliah.php

```php
<?php
class MataKuliah
{
    private $kode;
    private $nama;

    public function setData($kode, $nama)
    {
        $this->kode = $kode;
        $this->nama = $nama;
    }

    public function getData()
    {
        return [$this->kode, $this->nama];
    }
}
```

EXPLORER

UNTITLED (WORKSPACE)

lat1

Index.php

Mahasiswa.php

MataKuliah.php

Mahasiswa.php | MataKuliah.php | Index.php

lat1 > Index.php

```php
<?php
include "Mahasiswa.php";
include "MataKuliah.php";

$mahasiswa_1 = new Mahasiswa();
$makul_1 = new MataKuliah();
$makul_2 = new MataKuliah();

$makul_1->setData("A11.12345","Pemrograman Berbasis Web");
$makul_2->setData("A11.12346","Pemrograman Web Lanjut");

$mahasiswa_1->setData("A11.2024.15812", "Anthonius Raffael Pangaribuan", [$makul_1,$makul_2]);

echo "<pre>";
print_r($mahasiswa_1);
echo "</pre>";
```

OUTPUT

localhost/belajar-oop/lat1/

```
Mahasiswa Object
(
    [nim:Mahasiswa:private] => A11.2024.15812
    [nama:Mahasiswa:private] => Anthonius Raffael Pangaribuan
    [matkul:Mahasiswa:private] => Array
        (
            [0] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12345
                    [nama:MataKuliah:private] => Pemrograman Berbasis Web
                )

            [1] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12346
                    [nama:MataKuliah:private] => Pemrograman Web Lanjut
                )

        )

)
```

LATIHAN 2

EXPLORER

UNTITLED (WORKSPACE)
- lat1
  - Index.php
  - Mahasiswa.php
  - MataKuliah.php
- lat2
  - Index.php
  - Mahasiswa.php
  - MataKuliah.php

Mahasiswa.php | Index.php | MataKuliah.php

lat2 > Mahasiswa.php

```php
<?php
class Mahasiswa
{
    private $nim, $matkul;
    public $nama;

    public function setData($nim, $nama, $matkul)
    {
        $this->nim =$nim;
        $this->nama =$nama;
        $this->matkul =$matkul;
    }

    public function getData()
    {
        return[
            $this->nim,
            $this->nama,
            $this->matkul,
        ];
    }
}
```

```php
<?php
include "Mahasiswa.php";
include "MataKuliah.php";

$mahasiswa_1 = new Mahasiswa();
$makul_1 = new MataKuliah();
$makul_2 = new MataKuliah();

$makul_1->setData("A11.12345","Pemrograman Berbasis Web");
$makul_2->setData("A11.12346","Pemrograman Web Lanjut");

$mahasiswa_1->setData("A11.2024.15812", "Anthonius Raffael Pangaribuan", [$makul_1,$makul_2]);

echo "<pre>";
print_r($mahasiswa_1);
print_r($mahasiswa_1->nama);
print_r($mahasiswa_1->nim);
echo "</pre>";
```

OUTPUT

```
Mahasiswa Object
(
    [nim:Mahasiswa:private] => A11.2024.15812
    [matkul:Mahasiswa:private] => Array
        (
            [0] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12345
                    [nama:MataKuliah:private] => Pemrograman Berbasis Web
                )

            [1] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12346
                    [nama:MataKuliah:private] => Pemrograman Web Lanjut
                )

        )

    [nama] => Anthonius Raffael Pangaribuan
)
Anthonius Raffael Pangaribuan

Fatal error:  Uncaught Error: Cannot access private property Mahasiswa::$nim in C:\xampp\htdocs\Belajar-oop\lat2\Index.php:17
Stack trace:
#0 {main}
  thrown in C:\xampp\htdocs\Belajar-oop\lat2\Index.php on line 17
```

LATIHAN 3

EXPLORER

UNTITLED (WORKSPACE)
- lat1
- lat2
  - Index.php
  - Mahasiswa.php
  - MataKuliah.php
- lat3
  - Index.php
  - Mahasiswa.php
  - MataKuliah.php

Index.php | Mahasiswa.php | MataKuliah.php

lat3 > Index.php

```php
<?php
include "Mahasiswa.php";
include "MataKuliah.php";

$mahasiswa_1 = new Mahasiswa();
$makul_1 = new MataKuliah();
$makul_2 = new MataKuliah();

$makul_1->setData("A11.12345","Pemrograman Berbasis Web");
$makul_2->setData("A11.12346","Pemrograman Web Lanjut");

$mahasiswa_1->setData("A11.2024.15812", "Anthonius Raffael Pangaribuan", [$makul_1,$makul_2], "2004-12-27");

echo "<pre>";
echo "Umur dihitung dengan method/function : <br>" ;
print_r($mahasiswa_1->getData()[4]);
echo "<br><br>";
echo "Nilai Umur diberi secara langsung : <br>" ;
$mahasiswa_1->umur = 100;
print_r($mahasiswa_1->umur);
echo "</pre>";
```

EXPLORER

UNTITLED (WORKSPACE)
- lat1
- lat2
  - Index.php
  - Mahasiswa.php
  - MataKuliah.php
- lat3
  - Index.php
  - Mahasiswa.php
  - MataKuliah.php

Index.php | Mahasiswa.php | MataKuliah.php

lat3 > Mahasiswa.php

```php
<?php
class Mahasiswa
{
    private $nim, $nama, $matkul, $tgl_lahir;
    private $umur;

    public function setData($nim, $nama, $matkul, $tgl_lahir)
    {
        $this->nim =$nim;
        $this->nama =$nama;
        $this->matkul =$matkul;
        $this->tgl_lahir =$tgl_lahir;
        $this->setUmur();
    }

    public function setUmur()
    {
        $this->umur = date("Y") - substr($this->tgl_lahir, 0, 4);
    }

    public function getData()
    {
        return[
            $this->nim,
            $this->nama,
            $this->matkul,
            $this->tgl_lahir,
            $this->umur,
        ];
    }
}
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

No problems have been detected in the workspace.

OUTLINE

TIMELINE

```php
<?php
class MataKuliah
{
    private $kode;
    private $nama;

    public function setData($kode, $nama)
    {
        $this->kode = $kode;
        $this->nama = $nama;
    }

    public function getData()
    {
        return [$this->kode, $this->nama];
    }
}
```

```
Umur dihitung dengan method/function :
22

Nilai Umur diberi secara langsung :


Fatal error:  Uncaught Error: Cannot access private property Mahasiswa::$umur in C:\xampp\htdocs\Belajar-oop\lat3\Index.php:19
Stack trace:
#0 {main}
  thrown in C:\xampp\htdocs\Belajar-oop\lat3\Index.php on line 19
```

CONSTRUCT

```php
<?php
include "Mahasiswa.php";
include "MataKuliah.php";

$mahasiswa_1 = new Mahasiswa();
$makul_1 = new MataKuliah();
$makul_2 = new MataKuliah();

$makul_1->setData("A11.12345","Pemrograman Berbasis Web");
$makul_2->setData("A11.12346","Pemrograman Web Lanjut");

$mahasiswa_1->setData("A11.2024.15812", "Anthonius Raffael Pangaribuan", [$makul_1,$makul_2], "2004-12-27");

echo "<pre>";
print_r($mahasiswa_1->getData());
echo "</pre>";
```

```php
<?php
class Mahasiswa
{
    private $nim, $nama, $matkul, $tgl_lahir, $status;
    private $umur;

    public function __construct()
    {
        $this->status = "Aktif";
    }

    public function setData($nim, $nama, $matkul, $tgl_lahir)
    {
        $this->nim =$nim;
        $this->nama =$nama;
        $this->matkul =$matkul;
        $this->tgl_lahir =$tgl_lahir;
        $this->setUmur();
    }

    public function setUmur()
    {
        $this->umur = date("Y") - substr($this->tgl_lahir, 0, 4);
    }

    public function getData()
    {
        return[
            "nim" => $this->nim,
            "nama" => $this->nama,
            "matkul" => $this->matkul,
            "tgl_lahir" => $this->tgl_lahir,
            "umur" =>$this->umur,
            "status" =>$this->status,
        ];
    }
}
```

OUTPUT

```
Array
(
    [nim] => A11.2024.15812
    [nama] => Anthonius Raffael Pangaribuan
    [matkul] => Array
        (
            [0] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12345
                    [nama:MataKuliah:private] => Pemrograman Berbasis Web
                )

            [1] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12346
                    [nama:MataKuliah:private] => Pemrograman Web Lanjut
                )

        )

    [tgl_lahir] => 2004-12-27
    [umur] => 22
    [status] => Aktif
)
```

LATIHAN 4

EXPLORER

UNTITLED (WORKSPACE)
- lat1
- lat2
- lat3
- lat4
  - Index.php
  - Mahasiswa.php
  - MahasiswaTransfer.php
  - MataKuliah.php

Index.php | MahasiswaTransfer.php | MataKuliah.php | Mahasiswa.php

lat4 > Index.php

```php
<?php
include "Mahasiswa.php";
include "MataKuliah.php";
include "MahasiswaTransfer.php";

$makul_1 = new MataKuliah();
$makul_2 = new MataKuliah();

$daftar_nilai = [
    "1" => ["sks" => 2, "nilai" => "A"],
    "2" => ["sks" => 3, "nilai" => "B"],
    "3" => ["sks" => 2, "nilai" => "C"],
];

$mahasiswa_1 = new MahasiswaTransfer($daftar_nilai);

$makul_1->setData("A11.12345","Pemrograman Berbasis Web");
$makul_2->setData("A11.12346","Pemrograman Web Lanjut");

$mahasiswa_1->setData("A11.2024.15812", "Anthonius Raffael Pangaribuan", [$makul_1,$makul_2], "2004-12-27");

echo "<pre>";
print_r($mahasiswa_1->getData());
echo "</pre>";
```

EXPLORER

UNTITLED (WORKSPACE)
- lat1
- lat2
- lat3
- lat4
  - Index.php
  - Mahasiswa.php
  - MahasiswaTransfer.php
  - MataKuliah.php

OUTLINE
TIMELINE
VS CODE PETS

Index.php | MahasiswaTransfer.php | MataKuliah.php | Mahasiswa.php

lat4 > MahasiswaTransfer.php

```php
<?php
class MahasiswaTransfer extends Mahasiswa
{
    private $daftar_nilai, $ipk = 0, $ttl_sks = 0, $ttl_bobot = 0;
    private $bobot = ["A" => 4, "B" => 3, "C" => 2];

    public function __construct($daftar_nilai)
    {
        $this->daftar_nilai = $daftar_nilai;
        $this->getIPK();
        $this->status = 'Tidak Aktif';
    }

    public function getIPK()
    {
        if (!empty($this->daftar_nilai)) {
            foreach ($this->daftar_nilai as $data){
                $this->ttl_sks += $data['sks'];
                $this->ttl_bobot += $data['sks'] * $this->bobot[$data['nilai']];
            }

            $this->ipk = $this->ttl_bobot / $this->ttl_sks;
        }

        return $this->ipk;
    }

    public function getData()
    {
        return array_merge(parent::getData(),[
            "daftar_nilai" => $this->daftar_nilai,
            "ipk" => $this->ipk,
        ]);
    }
}
```

EXPLORER

UNTITLED (WORKSPACE)

- lat1
- lat2
- lat3
- lat4
  - Index.php
  - Mahasiswa.php
  - MahasiswaTransfer.php
  - MataKuliah.php

Index.php | MahasiswaTransfer.php | MataKuliah.php | Mahasiswa.php

lat4 > MataKuliah.php

```php
<?php
class MataKuliah
{
    private $kode;
    private $nama;

    public function setData($kode, $nama)
    {
        $this->kode = $kode;
        $this->nama = $nama;
    }

    public function getData()
    {
        return [$this->kode, $this->nama];
    }
}
```

EXPLORER

UNTITLED (WORKSPACE)

- lat1
- lat2
- lat3
- lat4
  - Index.php
  - Mahasiswa.php
  - MahasiswaTransfer.php
  - MataKuliah.php

Index.php | MahasiswaTransfer.php | MataKuliah.php | Mahasiswa.php

lat4 > Mahasiswa.php

```php
<?php
class Mahasiswa
{
    private $nim, $nama, $matkul, $tgl_lahir, $status;
    private $umur;

    public function __construct()
    {
        $this->status = "Aktif";
    }

    public function setData($nim, $nama, $matkul, $tgl_lahir)
    {
        $this->nim =$nim;
        $this->nama =$nama;
        $this->matkul =$matkul;
        $this->tgl_lahir =$tgl_lahir;
        $this->setUmur();
    }

    public function setUmur()
    {
        $this->umur = date("Y") - substr($this->tgl_lahir, 0, 4);
    }

    public function getData()
    {
        return[
            "nim" => $this->nim,
            "nama" => $this->nama,
            "matkul" => $this->matkul,
            "tgl_lahir" => $this->tgl_lahir,
            "umur" =>$this->umur,
            "status" =>$this->status,
        ];
    }
}
```

OUTLINE

TIMELINE

VS CODE PETS

Ln 4, Col 55 Spaces: 4

OUTPUT 1

(744) Aimer 『Ref:rain』MUSI

localhost/belajar-oop/lat4/

localhost/belajar-oop/lat4/

```
Array
(
    [nim] => A11.2024.15812
    [nama] => Anthonius Raffael Pangaribuan
    [matkul] => Array
        (
            [0] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12345
                    [nama:MataKuliah:private] => Pemrograman Berbasis Web
                )

            [1] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12346
                    [nama:MataKuliah:private] => Pemrograman Web Lanjut
                )

        )

    [tgl_lahir] => 2004-12-27
    [umur] => 22
    [status] =>
    [daftar_nilai] => Array
        (
            [1] => Array
                (
                    [sks] => 2
                    [nilai] => A
                )

            [2] => Array
                (
                    [sks] => 3
                    [nilai] => B
                )

            [3] => Array
                (
                    [sks] => 2
                    [nilai] => C
                )

        )

    [ipk] => 3
)
```

OUTPUT STATUS PROTECTED

(744) Aimer 『Ref:rain』MUSI

localhost/belajar-oop/lat4/

localhost/belajar-oop/lat4/

```
Array
(
    [nim] => A11.2024.15812
    [nama] => Anthonius Raffael Pangaribuan
    [matkul] => Array
        (
            [0] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12345
                    [nama:MataKuliah:private] => Pemrograman Berbasis Web
                )

            [1] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12346
                    [nama:MataKuliah:private] => Pemrograman Web Lanjut
                )

        )

    [tgl_lahir] => 2004-12-27
    [umur] => 22
    [status] => Tidak Aktif
    [daftar_nilai] => Array
        (
            [1] => Array
                (
                    [sks] => 2
                    [nilai] => A
                )

            [2] => Array
                (
                    [sks] => 3
                    [nilai] => B
                )

            [3] => Array
                (
                    [sks] => 2
                    [nilai] => C
                )

        )

    [ipk] => 3
)
```

EXPLORER

UNTITLED (WORKSPACE)

- lat1
- lat2
- lat3
- lat4
  - Index.php
  - Mahasiswa.php
  - MahasiswaTransfer.php
  - MataKuliah.php

OUTLINE

TIMELINE

VS CODE PETS

Index.php | MahasiswaTransfer.php | MataKuliah.php | Mahasiswa.php

lat4 > Mahasiswa.php

```php
<?php
class Mahasiswa
{
    private $nim, $nama, $matkul, $tgl_lahir;
    protected $status;
    private $umur;

    public function __construct()
    {
        $this->status = "Aktif";
    }

    public function setData($nim, $nama, $matkul, $tgl_lahir)
    {
        $this->nim =$nim;
        $this->nama =$nama;
        $this->matkul =$matkul;
        $this->tgl_lahir =$tgl_lahir;
        $this->setUmur();
    }

    public function setUmur()
    {
        $this->umur = date("Y") - substr($this->tgl_lahir, 0, 4);
    }

    public function getData()
    {
        return[
            "nim" => $this->nim,
            "nama" => $this->nama,
            "matkul" => $this->matkul,
            "tgl_lahir" => $this->tgl_lahir,
            "umur" =>$this->umur,
            "status" =>$this->status,
        ];
    }
```